

Highlights

- Demand uncertainty forces every online scheduler to a ratio of order $\sqrt{\theta}$.
- CARMA adds ghost jobs for expected arrivals to an exact min-cost-flow planner.
- CARMA meets deadlines, is θ -competitive, optimal with exact predictions.
- Dual prices give a core-stable carbon attribution among pooled data centres.
- On 2024 British grid data, CARMA cuts the gap to the optimum by a factor of 2-3.